Acute Infective Endocarditis. Highly virulent bacteria, most commonly *Staphylococcus aureus*, can colonize thrombi on a native (normal) valve, leading to large vegetations and rapid valve destruction. The high virulence of this bacterium also explains its acute onset. Symptoms of acute IE present rapidly, and the mortality rate is 50% despite treatment. Because the mitral or aortic valve may become quickly incompetent, acute heart failure (severe dyspnea, hypotension) can develop.

Subacute Infective Endocarditis. Less virulent bacteria, including the viridans streptococcal species *Streptococcus mutans* and *Streptococcus sanguinis*, cause smaller vegetations on damaged or genetically abnormal valves (e.g., bifid aortic valves or mitral valve prolapse). Symptoms thus present more insidiously. Subacute IE is less destructive, and most patients recover with treatment.

Prosthetic-Valve Endocarditis. Hardware within the heart provides a good environment for bacterial colonization. This includes artificial heart valves as well as other implantable devices like pacemaker wires and defibrillators. In particular, the slick texture of the prosthetic valve provides a good surface for biofilms or aggregates of bacteria surrounded by a sticky extracellular matrix. With time, age-related valve degeneration also provides new niduses for thrombus formation. The high-flow jets produced by the mechanical valve further damage the heart, while the adjacent low-pressure eddies promote thrombus formation.

Staphylococci most commonly colonize within 60 days of surgery. *S. epidermidis* is the most common, and *S. aureus* is the second most common. In contrast, streptococci most commonly colonize more than 60 days after surgery.

into the venous system. ***Pseudomonas aeruginosa*** is also common in these patients. Although not a part of our normal skin flora, *P. aeruginosa* can contaminate water and contaminate injections. After injection, the bacteria often lodge in the first valve they encounter—the **tricuspid valve**—making right-sided IE more common in this patient population.

Patients with Colon Cancer. These patients may have ulceration of the colon wall, which allows gut bacteria to enter the bloodstream. The gut flora ***Streptococcus gallolyticus*** (formerly *Streptococcus bovis*) is often implicated. If you have a patient diagnosed with *S. gallolyticus* endocarditis, consider that they may have an occult colon cancer.

Less Common Pathogens

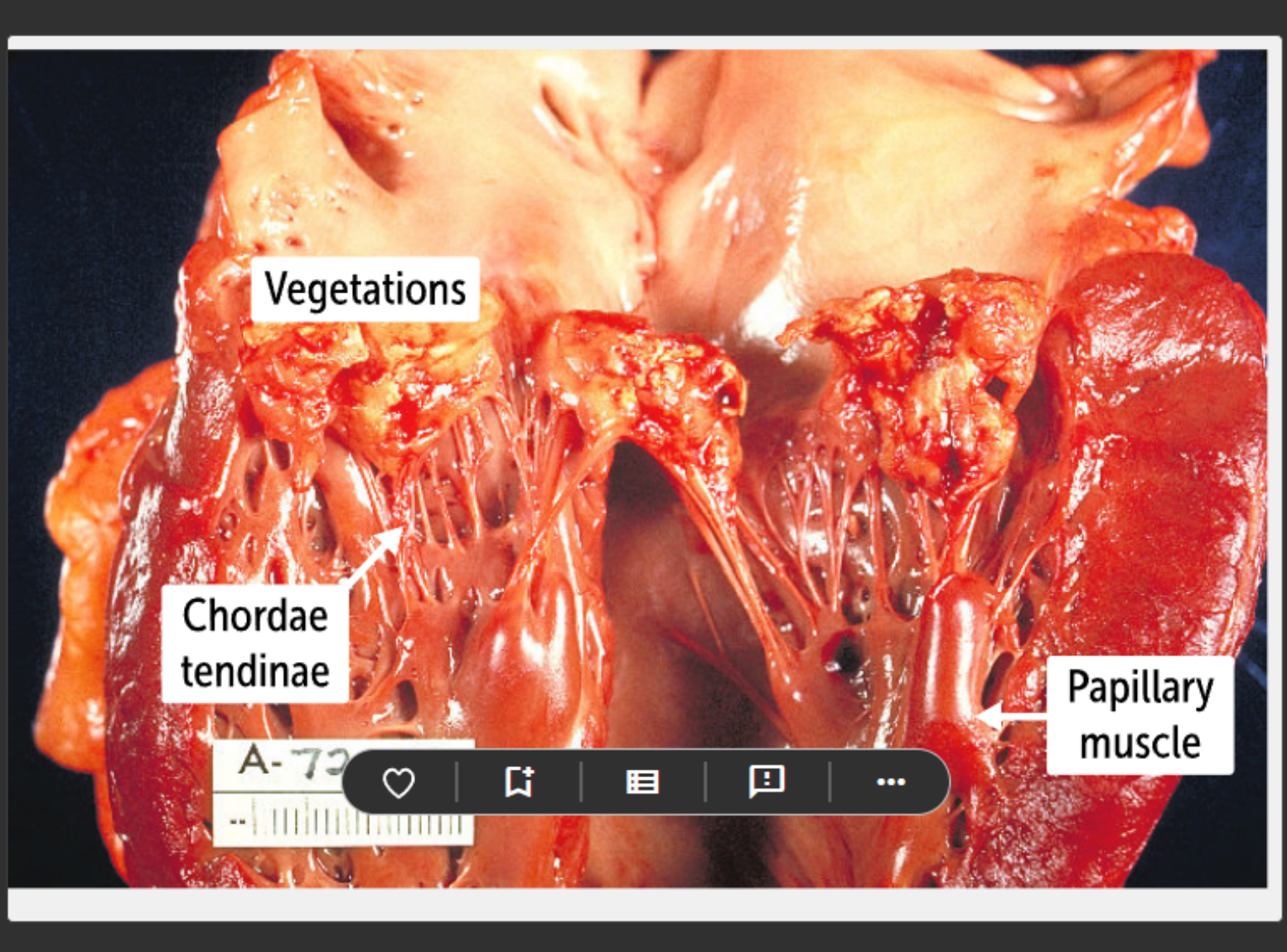
The **HACEK group of bacteria** are responsible for 5% to 10% of cases of IE in both native and prosthetic valves. HACEK stands for: *Haemophilus* species, *Aggregatibacter* species, *Cardiobacterium hominis*, *Eikenella corrodens*, and *Kingella* species. Because these are all fastidious **Gram-negative coccobacilli**, they grow slowly in culture; in fact, the cultures are often still negative after 3 days.

All can cause IE after dental procedures because they are part of the normal oropharyngeal flora. Of these organisms, *Eikenella* is most often associated with IV injectable drug use, while *Kingella* is known for causing rapid progression.

Fungi can also cause serious infections, especially in immunocompromised patients, patients who use injectable drugs, and postsurgical patients (any type of surgery). It's unco *Candida* and *Aspergillus* species are the most commonly implicated. *Candida* can form in IE (the left heart has higher intrachamber pressures). The reason for this is twofold:

- First, high-pressure jet streams create adjacent low-flow eddies, which are the perfect focus for clot formation along the valves. With the introduction of bacteria, the valves easily become infected.
- Second, high-flow valves are more likely to be damaged, offering a nidus for additional thrombosis.

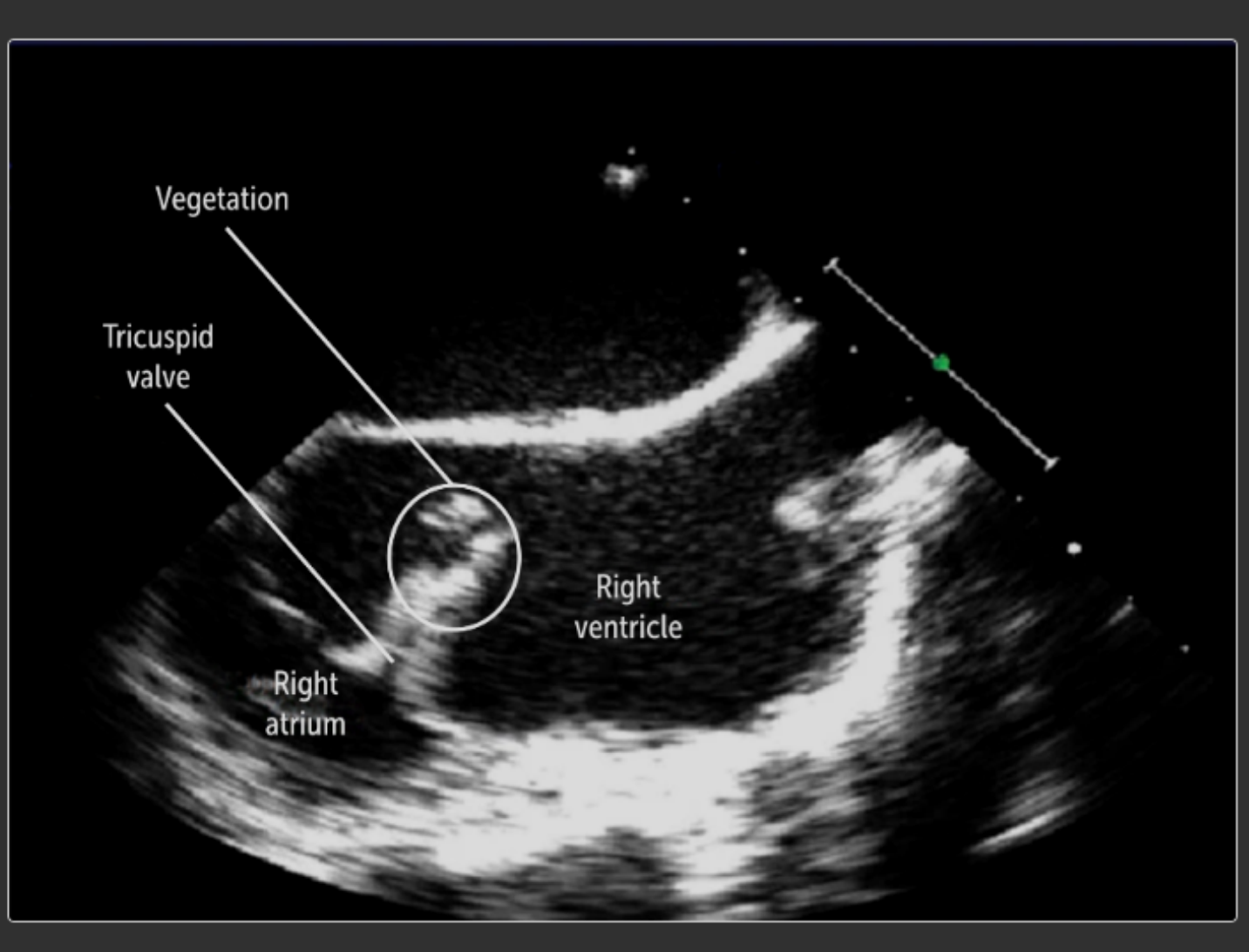
In the general population, **mitral regurgitation** is a common valvular abnormality. Figure 1 shows vegetations involving the mitral valve that developed as a result of valvular damage. These clots are then colonized by bacteria that are circulating in the blood.



QUIZ

Figure 1. Mitral Valve Vegetations.

In contrast, in patients who use injectable drugs, the **tricuspid valve** is the most frequently involved valve (Figure 2).



ALICE L WALTON SCHOOL OF MEDICINE

Mitral Stenosis

Last updated February 20, 2026 17 min read CC BY-SA 4.0

Adapted by: Diptiman Bose, Patrick Brooks, Phil Riley, Ian Murray

Editors: Chelsey Deel, Jordan Finn, Russell Johnson

Adapted from: Mitral Stenosis

Listen to this Brick
0:00 / 26:02 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Define mitral stenosis and its clinical presentation.
2. Explain the etiology, pathophysiology, and complications of mitral stenosis.
3. Explain the diagnosis of mitral stenosis.
4. Describe the

What do you want to learn?

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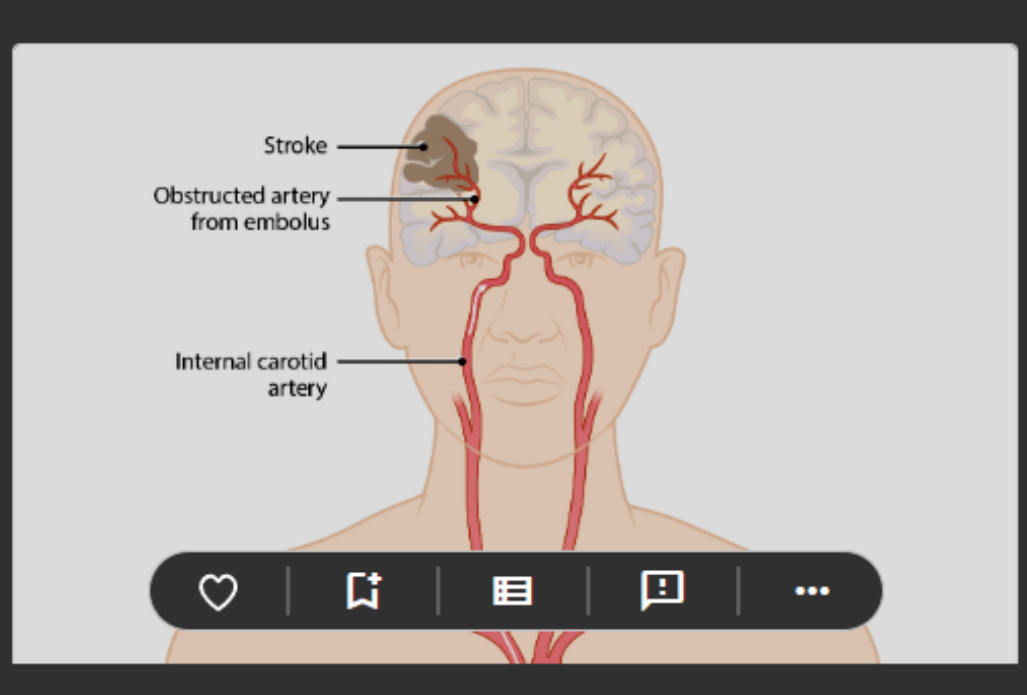
Complications of Mitral Stenosis (LO 2)

Most complications of MS are related to structural changes to the LA that lead to atrial fibrillation (AF) and thromboembolism.

Atrial Fibrillation (A fib). MS (and MR) may cause A fib, an irregularly irregular heart rhythm, in 30%-70% of patients. This is due to enlargement of the LA, which dilates to accommodate the extra blood volume. As the LA expands, the normal architecture of the heart wall is stretched, disrupting its electrical activity and increasing the risk of A fib.

Thromboembolism. In atrial dilation and A fib, the lack of regular atrial contractions allows blood to remain static in the LA. This increases the risk of formation of blood clots (atrial thrombi). This can lead to thromboembolism, in which a blood clot travels downstream from the LA and exits the heart to occlude the arterial blood supply. This occlusion occurs most often in a part of the cerebral vasculature, resulting in a **cardioembolic stroke** (Figure 7).



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4. Describe the

This leads to tricuspid regurgitation, which can cause a bounding jugular venous pulse during systole (as the right ventricular contraction is transmitted into the right atrium and jugular veins).

Why is the tricuspid valve most involved in IE in this setting? Injected drugs may be contaminated with compounds capable of inducing endothelial damage. When a needle is inserted into a vein, these products, along with any skin flora or bacteria inside the syringe or mixed with the drug, will be injected into venous blood, and when these products and bacteria reach the heart, the tricuspid valve is the first valve they encounter.

Why are people who use injectable drugs more prone to infective endocarditis (IE) of the tricuspid valve?

Complications

B. ○ Shortness of breath and lower extremity swelling

C. ● Fever and new onset heart murmur

D. ○ Joint and back pain

E. ○ Fatigue and lethargy

Hide Explanation

The correct answer is fever and new onset heart murmur (C). Fever is present in up to 90% of patients, and a new heart murmur in 85% of patients. This combination of symptoms would be very helpful in identifying a patient with IE. Altered mental status and vision changes (A), shortness of breath and lower extremity swelling (B), joint and back pains (D), and fatigue and lethargy (E) are all signs and symptoms that can be seen with IE, but these symptoms vary and are commonly absent.

5. Which imaging modality is most commonly used in the diagnosis of IE?

A. ○ Chest x-ray

B. ○ Computed tomography scan

C. ○ Magnetic resonance imaging

D. ● Echocard

ALICE L WALTON SCHOOL OF MEDICINE

Infective Endocarditis

Last updated March 9, 2026 20 min read CC BY-SA 4.0

Adapted by: Rhome Hughes, Chelsey Deel, Stephen Nix, Diptiman Bose, Ray Brown, Lori Garman

Editors: Jordan Finn, Russell Johnson

Adapted from: Infective Endocarditis

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Learning Objectives (5) Versions

After completing this brick, you will be able to:

1. Recognize the three types of infective endocarditis (IE).
2. Differentiate the clinical presentation of IE.
3. Explain the pathophysiology of IE.
4. Interpret clinical findings to properly diagnose IE.
5. Outline the